

Project Programming

title: "Energy poverty in the Netherlands" author: "Matthijs, David, Wessel, Quint, Zakaria" date: "2026-06-08" output: pdf_document: default html_document: default —

Part 1: Identify a Social Problem

1.1 Describe the Social Problem

In recent years, energy prices in the Netherlands have increased significantly, leading to a substantial rise in fixed costs of Dutch households. This increase in fixed expenses reduces the financial resources available for those household, limiting their capacity to meet other essential needs. To effectively address this growing issue among those households, it is crucial to quantify and understand the extent to which rising energy prices drive poverty rates among the Dutch households.

Part 2: Data Sourcing

2.1 Load in the data

```
raw_energie <- read_csv2("Energietarieven.csv")
```

```
## i Using ',', ' as decimal and '.' as grouping mark. Use `read_delim()` for more control.
```

```
## Rows: 3 Columns: 11
```

```
## -- Column specification -----
```

```
## Delimiter: ";"
```

```
## dbl (9): Jaar, Aardgas transporttarief inc BTW Euro/jaar, Aardgas leverings ...
```

```
## lgl (2): Elektriciteit|Vast leveringstarief dynamisch inc btwEuro/jaar, Elek...
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
raw_inkomen <- read_csv2("Inkomen huishoudens samengevoegd.csv")
```

```
## i Using ',', ' as decimal and '.' as grouping mark. Use `read_delim()` for more control.
```

```
## Rows: 51 Columns: 8-- Column specification -----
```

```
## Delimiter: ";"
```

```
## chr (1): Regios
```

```
## dbl (7): Jaar, Particuliere huishoudens x 1 000, Particuliere huishoudens re...
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
dataset_samengevoegd <- raw_inkomen %>%
  left_join(raw_energie, by = "Jaar")
colnames(raw_energie)
```

```
## [1] "Jaar"
## [2] "Aardgas transporttarief inc BTW Euro/jaar"
## [3] "Aardgas leverings tarief vast inc btw Euro/jaar"
## [4] "Aardgas|Variabel leveringstarief inc btw Euro/m3"
## [5] "aardgas energiebelasting inc btw/m3"
## [6] "elektriciteit transporttarief inc btwEuro/jaar"
## [7] "elek vast leveringstarief inc btwEuro/jaar"
## [8] "Elektriciteit|Variabel leveringstarief contractprijs inc btwEuro/kWh"
## [9] "Elektriciteit|Vast leveringstarief dynamisch inc btwEuro/jaar"
## [10] "Elektriciteit|Variabel leveringstarief dynamisch inc btwEuro/kWh"
## [11] "Elektriciteit|Energiebelasting inc btwEuro/kWh"
```

```
colnames(raw_inkomen)
```

```
## [1] "Jaar"
## [2] "Regios"
## [3] "Particuliere huishoudens x 1 000"
## [4] "Particuliere huishoudens relatief%"
## [5] "Inkomen|Gemiddeld gestandaardiseerd inkomen 1 000 euro"
## [6] "Inkomen|Mediaan gestandaardiseerd inkomen 1 000 euro"
## [7] "Inkomen|Gemiddeld besteedbaar inkomen 1 000 euro"
## [8] "Inkomen|Mediaan besteedbaar inkomen 1 000 euro"
```

2.2 Short summary of the data sets

The data sets utilized in this study consist of official statistics sourced from Statistics Netherlands (CBS). The first data set contains average consumer energy tariffs, while the second data set comprises Dutch household income levels across the years 2021, 2022, and 2023, broken down by province (CBS, 2023).

A limitation of the first data set is that it does not capture the exact monthly or annual energy usage of individual households. However, it provides a reliable overview of the national average fixed and variable tariffs for both natural gas and electricity.

The second data set includes detailed information on standardized, median, and disposable household incomes. Utilizing these specific metrics ensures the most reliable and realistic representation of household financial capacity across the different Dutch provinces (CBS, 2023).

Part 3: Quantifying

Data cleaning

During the import of the Excel tables from the CBS website, it was found that the names of the columns did not match. The name of the first column in the Income datasheet was “jaar”, while the name of the first column in the Energie datasheet was “perioden”. That issue caused an error in R, it was solved by naming both first columns “jaar”. The energy data set was also cleaned by deleting the loose months and use the years instead. After these adjustments the data sets were merged by using `left_join()`, and R could read the files correctly.

Describing variabels

Jaar: The temporal identifier factor (2021, 2022, 2023).

Regios: The geographic identifier containing both provincial jurisdictions and major municipalities.

income_median: The median disposable household income expressed in thousands of Euros.

energy_burden: The primary composite index representing the percentage of median disposable income spent on combined baseline energy costs. # Creation of the new variables

```
# We berekenen de variabelen én hernoemen de oude kolommen in één stap!
dataset_berekend <- dataset_samengevoegd %>%
  mutate(
    # 1. We geven het inkomen meteen een kortere Engelse naam
    income_median = `Inkomen|Mediaan besteedbaar inkomen 1 000 euro`,

    # 2. Total gas costs calc (met de originele CBS-namen)
    gas_costs = `Aardgas transporttarief inc BTW Euro/jaar` +
      `Aardgas leverings tarief vast inc btw Euro/jaar` +
      (1200 * (`Aardgas|Variabel leveringstarief inc btw Euro/m3` + `aardgas energieleveringstarief inc btw Euro/m3`)),

    # 3. Total electricity costs calc (met de originele CBS-namen)
    electricity_costs = `elektriciteit transporttarief inc btwEuro/jaar` +
      `elek vast leveringstarief inc btwEuro/jaar` +
      (2500 * (`Elektriciteit|Variabel leveringstarief contractprijs inc btwEuro/kWh` + `elektriciteit contractprijs inc btwEuro/kWh`)),

    # 4. Total energy costs combined
    total_energy_costs = gas_costs + electricity_costs,

    # 5. energy_burden (totale energiekosten)
    energy_burden = (total_energy_costs / (income_median * 1000)) * 100,

    # 6. gas_burden (pure gaskosten)
    gas_burden = (gas_costs / (income_median * 1000)) * 100,

    # 7. electricity_burden (pure stroomkosten)
    electricity_burden = (electricity_costs / (income_median * 1000)) * 100
  )
```

Explanation new variables

The new variables for this study are:

Energy_burden, represents the portion of the Dutch median disposable income that is spent on total energy costs (gas and electricity combined).

Gas_burden, represents the portion of the Dutch median disposable income that is spent on gas costs. This variable is calculated by adding the variable and fixed costs of gas based on a consumption of 1200m3 per year, the CBS guidelines show that 1200m3 is the normal consumption for a household per year.

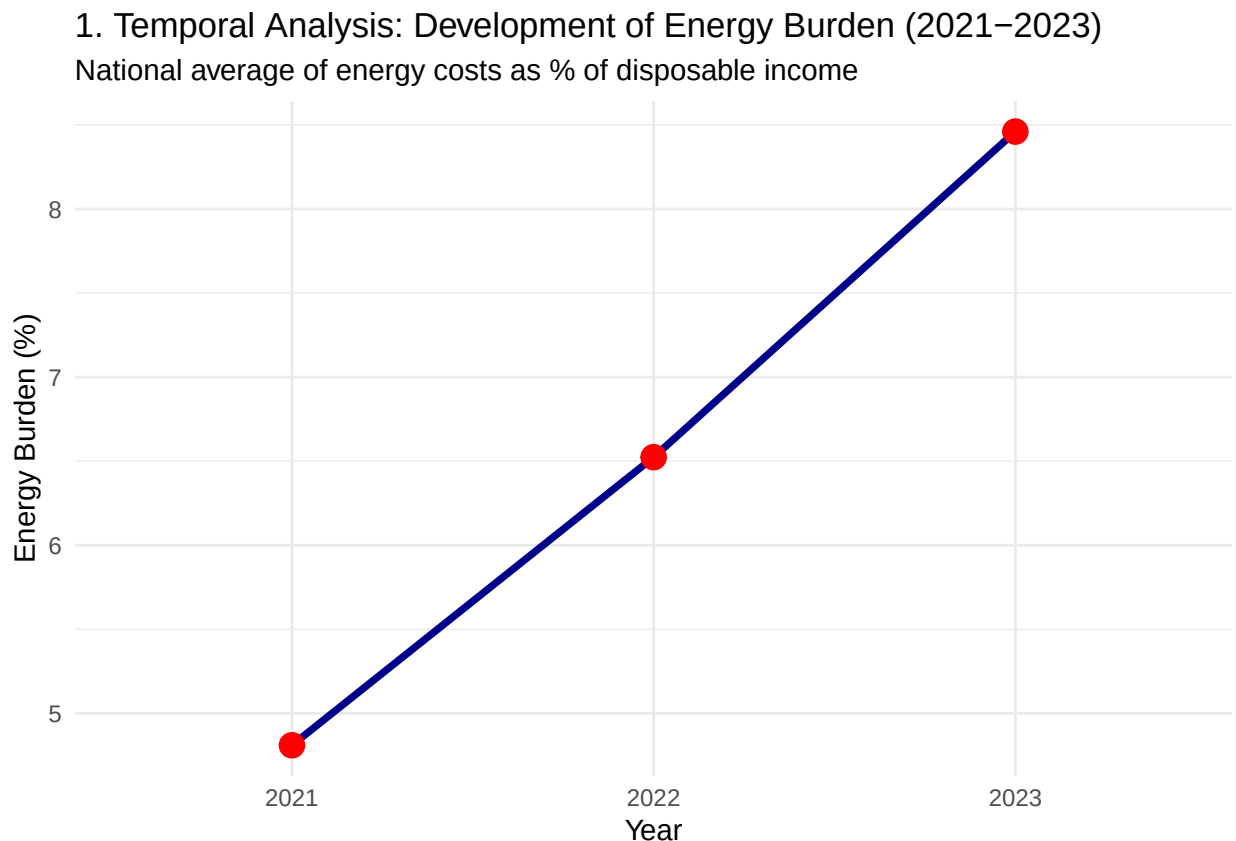
Electricity_burden, represents the portion of the Dutch median disposable income that is spent on electricity costs. This variable is calculated by adding the variable and fixed costs of electricity based on a consumption of 2500kWh per year, the CBS guidelines show that 2500kWh is the normal consumption for a household per year(CBS,2023).

Data Visualisation

Figure 1 Temporal Variation

```
# 1. Data preperation
temporal_data <- dataset_berekend %>%
  filter(Regios == "Nederland")

ggplot(temporal_data, aes(x = factor(Jaar), y = energy_burden, group = 1)) +
  geom_line(color = "darkblue", linewidth = 1.3) +
  geom_point(color = "red", size = 4) +
  labs(
    title = "1. Temporal Analysis: Development of Energy Burden (2021-2023)",
    subtitle = "National average of energy costs as % of disposable income",
    x = "Year",
    y = "Energy Burden (%)"
  ) +
  theme_minimal()
```



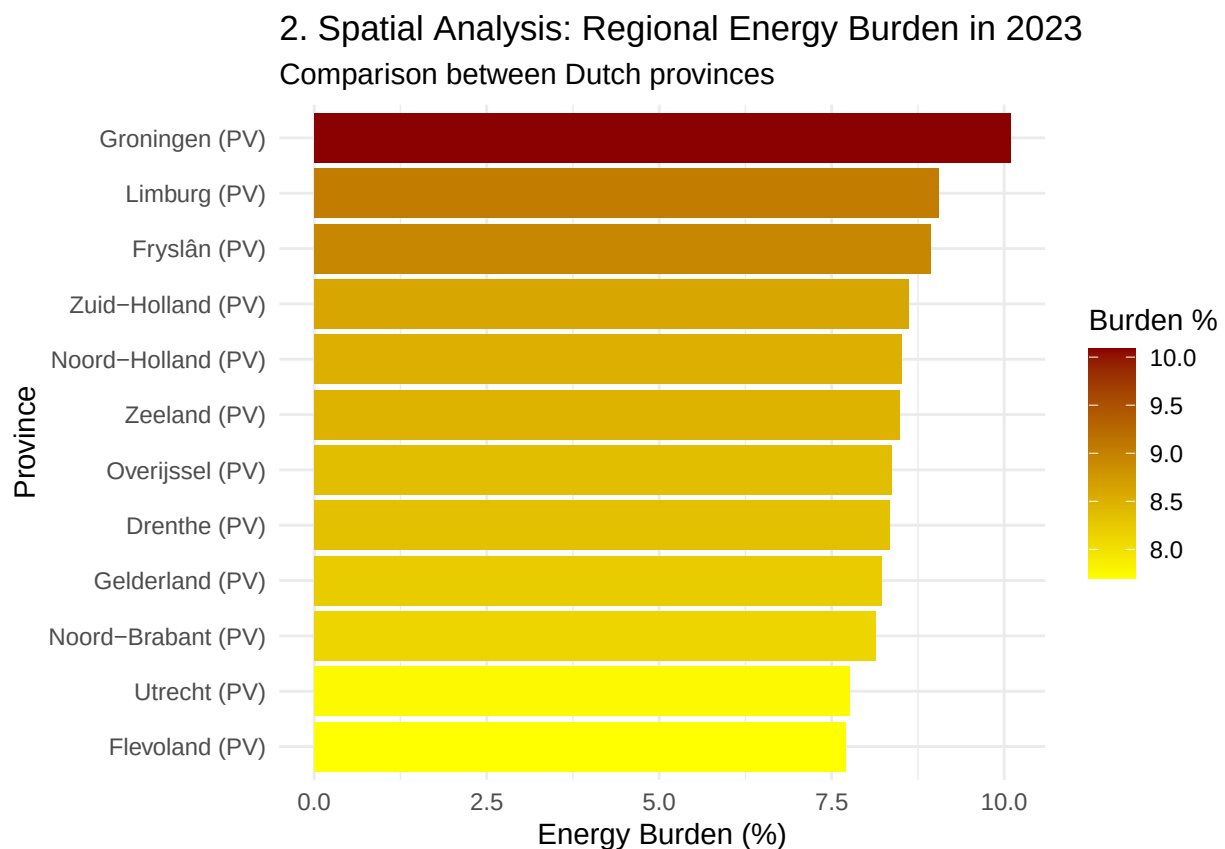
This graph shows that the raw energy prices rise significantly and rise in percentage of the household income. The data only shows this raw energy price, because it is important to see what happens if the government does not intervene. #Figure 2 Spatial Variation

```

spatial_data <- dataset_berekend %>%
  filter(Jaar == 2023 & grepl("\\(PV\\)", Regios))

ggplot(spatial_data, aes(x = reorder(Regios, energy_burden), y = energy_burden, fill = energy_burden)) +
  geom_col() +
  coord_flip() + # Horizontaal maken voor leesbaarheid regio's
  scale_fill_gradient(low = "yellow", high = "darkred") +
  labs(
    title = "2. Spatial Analysis: Regional Energy Burden in 2023",
    subtitle = "Comparison between Dutch provinces",
    x = "Province",
    y = "Energy Burden (%)",
    fill = "Burden %"
  ) +
  theme_minimal() +
  theme(legend.position = "right")

```



This graph shows that there is a significant difference between the region in the Netherlands. #Figure 3
Sub Population Variation

```

sub_population_data <- dataset_berekend %>%
  filter(Jaar == 2023) %>%
  mutate(Type = case_when(
    Regios == "Nederland" ~ "National Average",
    grepl("\\(PV\\)", Regios) ~ "Provinces",
    TRUE ~ "Major Cities" # De rest (zoals Amsterdam) zijn steden
  ))

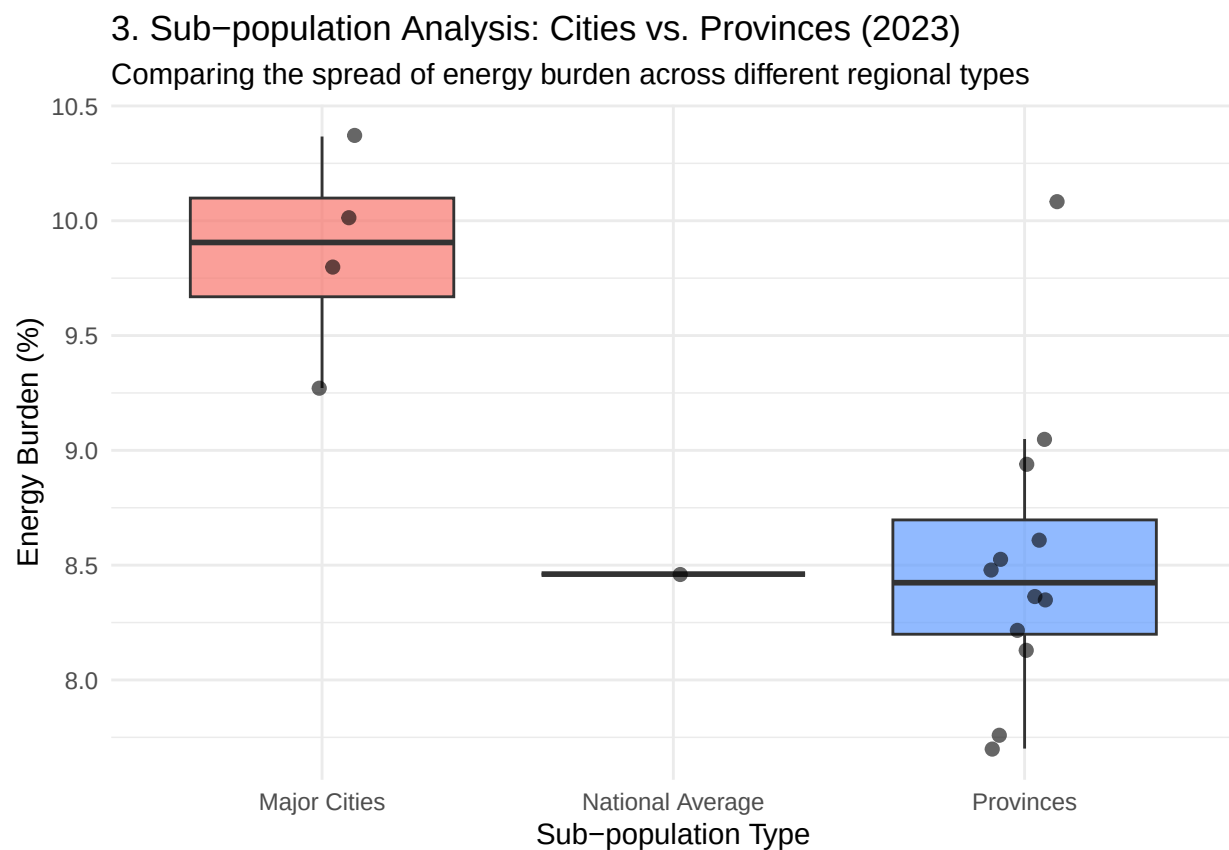
```

```

))

ggplot(sub_population_data, aes(x = Type, y = energy_burden, fill = Type)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA) +
  geom_jitter(width = 0.1, size = 2, alpha = 0.6) +
  labs(
    title = "3. Sub-population Analysis: Cities vs. Provinces (2023)",
    subtitle = "Comparing the spread of energy burden across different regional types",
    x = "Sub-population Type",
    y = "Energy Burden (%)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")

```



#Figure 4 Event Analysis

```

event_data <- dataset_berekend %>%
  filter(Regios != "Nederland")

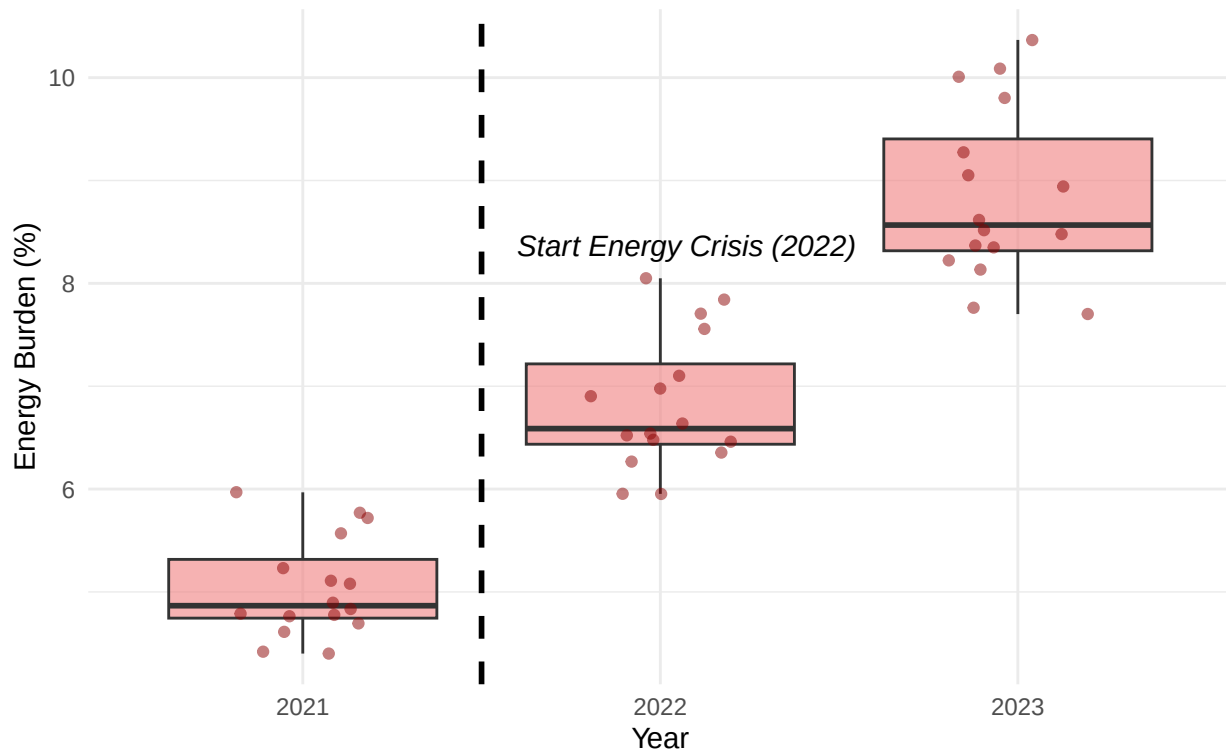
ggplot(event_data, aes(x = factor(Jaar), y = energy_burden)) +
  geom_boxplot(fill = "lightcoral", alpha = 0.6, outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.5, color = "darkred") +
  # Voeg een visuele lijn/indicator toe die het 'Event' markeert (de start van de energiecrisis begin 2022)
  geom_vline(xintercept = 1.5, linetype = "dashed", color = "black", linewidth = 1) +
  annotate("text", x = 1.6, y = max(event_data$energy_burden) - 2,
    label = "Start Energy Crisis (2022)", hjust = 0, fontface = "italic", color = "black") +

```

```
labs(
  title = "4. Event Analysis: Impact of the 2022 Energy Crisis",
  subtitle = "Shift in household energy burden before and after the price shock",
  x = "Year",
  y = "Energy Burden (%)"
) +
theme_minimal()
```

4. Event Analysis: Impact of the 2022 Energy Crisis

Shift in household energy burden before and after the price shock



Once the war started in Ukraine in 2022 the energy prices for Dutch households rose. # Discussion

Our findings state that, within the limitations mentioned earlier, the share of the Dutch median income spent on energy bills increased significantly each year between 2021 and 2023. It is important to note that precise figures are difficult to determine. However, the fact remains that standard disposable income has not increased as rapidly as energy costs. This trend highlights the growing importance of policies aimed at improving energy affordability and protecting vulnerable households in the Netherlands. While our raw tariff data shows a structural increase in energy burdens, TNO reports that the actual number of households in official 'energy poverty' decreased in 2023 from 520,000 to 450,000 (Mulder et al., 2023, p. 3). This deviation is explained by aggressive, temporary government interventions, specifically the national price cap (prijssplafond) and the €1,300 energy allowance for low-income sub-populations. However this study shows that the raw energy tariffs for Dutch households rise significantly and will be a problem if the government does not intervene.

Reproductibility

How this study is produced:

Project_programming.Rmd/

README.md # Project onboarding, data sources, and run instructions Project_programming.Rmd #
The primary executable Rmarkdown analysis report data/ Energietarieven.csv # Raw CBS energy
tariffs data Inkomen huishoudens samengevoegd.csv # Raw CBS regional income data

We downloaded multiple data sets from the CBS website for our data we made a average usage based on
the dataset gemiddeld verbruik per huishouden. For the other datasets they are in the references.

References

Mulder et al., 2023, p. 3 <https://publications.tno.nl/publication/34642676/7qBbU9FJ/TNO-2024-R10801.pdf>
CBS Gemiddelde energietarieven voor consumenten <https://www.cbs.nl/nl-nl/cijfers/detail/85592NED>
CBS Inkomen van huishoudens; huishoudenskenmerken, regio (indeling 2024) Nederland <https://www.cbs.nl/nl-nl/cijfers/detail/86004NED>
CBS Gemiddelde verbruik per huishouden <https://www.cbs.nl/nl-nl/cijfers/detail/81528NED>